Section 3 : Networking Basics

Task 3.1: Network Concepts

1. **Explain the OSI Model**

The OSI (Open Systems Interconnection) Model is a framework that explains how data is transmitted over a network in 7 layers.

- List all 7 layers

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

- Give one example protocol for each layer

1. Physical Layer -> Ethernet
2. Data Link Layer -> MAC
3. Network Layer -> IP
4. Transport Layer -> TCP
5. Session Layer -> NetBIOS
6. Presentation Layer -> SSL
7. Application Layer -> HTTP

- Explain which layer TCP operates at

TCP operates at the Transport Layer (Layer 4) because this layer is responsible for end-to-end communication between devices.

The Transport Layer ensures that:

- Data is delivered correctly
- Data is delivered in the right order
- Missing data is resent if lost

2. **IP Addressing**
   - What is the difference between public and private IP?

Feature	Public IP	Private IP
Usage	Used on the internet	Used inside local networks
Uniqueness	Globally unique	Not unique (can repeat)
Accessibility	Accessible from anywhere	Not directly accessible from internet
Assignment	Assigned by ISP	Assigned by router
Security	Less secure (public exposure)	More secure (internal use)
Example	8.8.8.8	192.168.1.100

- List the three private IP ranges (Class A, B, C)

1. **Class A:** 10.0.0.0 – 10.255.255.255
2. **Class B:** 172.16.0.0 – 172.31.255.255
3. **Class C:** 192.168.0.0 – 192.168.255.255

- Is 192.168.1.100 a public or private IP?

Private IP address - Because it belongs to the Class C private range

Task 3.2: Practical Network Analysis

1. **Display your network configuration**

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.105  netmask 255.255.255.0  broadcast 192.168.0.255
        inet6 fe80::710c:7719:43d8:886c  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:6d:4a:ce  txqueuelen 1000  (Ethernet)
        RX packets 282  bytes 18238 (17.8 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 65  bytes 6322 (6.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Command used : "ifconfig"

Explanation

The **ifconfig** command displays network configuration details such as IP address and interface information. System IP address is 192.168.0.105.

2. **Show your routing table**

```
┌──(kali㉿kali)-[~]
└─$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.0.1     0.0.0.0         UG    100    0        0 eth0
192.168.0.0     0.0.0.0         255.255.255.0   U     100    0        0 eth0
```

Command used : "route -n"

Explanation

The **route -n** command displays the routing table of the system. It shows how network traffic is routed. The default gateway is **192.168.0.1**, which is used to send data from the local network to the internet.

3. **Display active network connections**

```
┌──(kali㉿kali)-[~]
└─$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
```

Command used : "netstat -tuln"

Explanation

The **netstat -tuln** command displays active network connections and listening ports

on the system. In this output, no active listening ports are shown, which means no services are currently running.

4. Test connectivity to google.com (ping)

```
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=202 ttl=116 time=42.5 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=203 ttl=116 time=11.5 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=204 ttl=116 time=12.0 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=205 ttl=116 time=24.5 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=206 ttl=116 time=11.8 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=207 ttl=116 time=12.5 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=208 ttl=116 time=10.5 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=209 ttl=116 time=16.6 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=210 ttl=116 time=11.0 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=211 ttl=116 time=105 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=212 ttl=116 time=49.3 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=213 ttl=116 time=21.2 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=214 ttl=116 time=9.82 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=215 ttl=116 time=10.7 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=216 ttl=116 time=11.6 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=217 ttl=116 time=11.1 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=218 ttl=116 time=9.43 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=219 ttl=116 time=10.0 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=220 ttl=116 time=18.7 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=221 ttl=116 time=26.9 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=222 ttl=116 time=9.38 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=223 ttl=116 time=9.96 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=224 ttl=116 time=11.8 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=225 ttl=116 time=10.9 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=226 ttl=116 time=9.90 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=227 ttl=116 time=13.0 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=228 ttl=116 time=15.1 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=229 ttl=116 time=11.0 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=230 ttl=116 time=10.8 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=231 ttl=116 time=9.37 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=232 ttl=116 time=9.42 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=233 ttl=116 time=10.3 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=234 ttl=116 time=9.35 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=235 ttl=116 time=10.0 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=236 ttl=116 time=12.2 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=237 ttl=116 time=9.74 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=238 ttl=116 time=9.72 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=239 ttl=116 time=10.5 ms
64 bytes from lcmaaa-ao-in-f14.1e100.net (142.251.223.14): icmp_seq=240 ttl=116 time=13.2 ms
^C
--- google.com ping statistics ---
240 packets transmitted, 240 received, 0% packet loss, time 239594ms
rtt min/avg/max/mdev = 9.353/22.065/229.818/24.526 ms
```

Command used: "ping google.com"

Explanation:

The ping command is used to test connectivity between my system and google.com. The continuous replies show that my system is successfully connected to the internet. The output also displays the time taken for each packet, indicating network performance.

5. Trace route to google.com

```
┌──(kali㉿kali)-[~]
└─$ traceroute google.com
traceroute to google.com (142.251.223.14), 30 hops max, 60 byte packets
 1  192.168.0.1 (192.168.0.1)  4.610 ms  4.454 ms  4.392 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  49.205.72.39.actcorp.in (49.205.72.39)  9.873 ms * *
 7  72.14.243.242 (72.14.243.242)  106.619 ms  106.545 ms  8.566 ms
 8  * * *
 9  142.250.235.104 (142.250.235.104)  11.361 ms 74.125.253.12 (74.125.253.12)  11.158 ms 74.125.253.16 (74.125.253.16)  11.408 ms
10  142.251.55.43 (142.251.55.43)  11.257 ms 172.253.75.14 (172.253.75.14)  11.021 ms 142.251.55.41 (142.251.55.41)  10.905 ms
11  142.250.208.231 (142.250.208.231)  11.643 ms 142.250.63.173 (142.250.63.173)  11.491 ms 142.251.51.119 (142.251.51.119)  10.240 ms
12  142.251.55.41 (142.251.55.41)  10.123 ms cgk02s06-in-f14.1e100.net (142.251.223.14)  8.061 ms 142.251.55.41 (142.251.55.41)  8.491 ms
```

Command used: traceroute google.com

Explanation:

The traceroute command shows the path taken by data packets from my system to google.com. It displays each router (hop) along the route. In this output, the total number of hops to reach google.com is **12**.